

PhD Math Camp Exam 2023

Columbia University

August 28, 2023
09:00am

Abstract

This is a timed remote exam. You have time until noon (12pm) to submit your answers via email to the TA (aza2115@columbia.edu) and cc the instructor (mnb2141@columbia.edu). There are eight questions for a total of 120 points (plus one bonus question for an additional 10 points). Please justify your answers, partial credit will be awarded in case of incomplete answers. The exam is closed book and notes, and you are not allowed to consult peers or use the web to aid in solving the problems. A non-programmable calculator is allowed. Columbia's academic honor code applies.

1. (15 points) Answer whether each is a distance in $\mathbb{R}^2$ and justify your answer.

(a) $d(x, y) := \min \{|x_1 - y_1|, |x_2 - y_2|\}$

Answer: No. $d((0, 0), (0, 1)) = 0$ despite $(0, 0) \neq (0, 1)$.

(b) $d(x, y) := (\|x\| + \|y\|) \cdot \mathbf{1}_{x \neq y}$, where $\|\cdot\|$ is the Euclidean norm (i.e. $\|z\| := \sqrt{z_1^2 + z_2^2}$) and $\mathbf{1}_{x \neq y} = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$

Answer: Yes. It is trivial to check that $d(x, y) \geq 0$, $d(x, y) = 0 \iff x = y$, and that $d(x, y) = d(y, x) \quad \forall x, y$.

Triangle inequality says $\forall x, y, z \in X$, $(\|x\| + \|y\|)\mathbf{1}_{x \neq y} + (\|y\| + \|z\|)\mathbf{1}_{y \neq z} \geq (\|x\| + \|z\|)\mathbf{1}_{x \neq z}$, which holds with equality if $x = y$ and/or $y = z$; otherwise, the LHS is $\|x\| + 2\|y\| + \|z\|$, which is clearly at least as great as the RHS.

2. (15 points) Check whether the following sets are subspaces of the n -dimensional real vector space $\langle \mathbb{R}^n, +, \cdot \rangle$, where vector addition and scalar multiplication are defined in the naive component-by-component fashion.

(a) When $n = 1$, the set of integers $\mathbb{Z}$.

Answer: No. For $1 \in \mathbb{Z}$, take $\lambda = 0.5$, then $\lambda \cdot 1 = 0.5 \notin \mathbb{Z}$

(b) $\text{Ker } A := \{\mathbf{v} \in \mathbb{R}^n : A\mathbf{v} = \mathbf{0}\}$, where A is an $n \times n$ real matrix.

Answer: Yes. Notice that $\mathbf{0} \in \text{Ker } A$ and that for any $\mathbf{v}, \mathbf{w} \in \text{Ker } A$ and $\lambda, \mu \in \mathbb{R}$, we have $A(\lambda\mathbf{v} + \mu\mathbf{w}) = \lambda A\mathbf{v} + \mu A\mathbf{w} = \lambda \cdot \mathbf{0} + \mu \cdot \mathbf{0} = \mathbf{0}$. Thus, $\lambda\mathbf{v} + \mu\mathbf{w} \in \text{Ker } A$.

3. (15 points) For each of the following functions, show whether the function is homogeneous (if so, state its degree of homogeneity), and state whether it is homothetic and why. If your answer depends on the parameters, explain how it depends on the parameters.

(a) $u(x_1, x_2, \dots, x_n) = (\sum_{i=1}^n a_i x_i^\gamma)^{1/\gamma}$.

(b) $u(x_1, x_2, \dots, x_n) = a_0 + a_1 x_1 + \dots + a_n x_n$.

Answer:

(a) For any set of parameters, the function is homogeneous of degree 1 and therefore also homothetic.

To see this, note that

$$\begin{aligned} u(\lambda x_1, \lambda x_2, \dots, \lambda x_n) &= \left(\sum_{i=1}^n a_i (\lambda x_i)^\gamma \right)^{1/\gamma} \\ &= \left(\lambda^\gamma \sum_{i=1}^n a_i x_i^\gamma \right)^{1/\gamma} \\ &= \lambda \left(\sum_{i=1}^n a_i x_i^\gamma \right)^{1/\gamma} \\ &= \lambda u(x_1, x_2, \dots, x_n) \end{aligned}$$

(b) If $a_0 = 0$, u is homogeneous of degree 1 and therefore also homothetic. To see this, note that

$$\begin{aligned} u(\lambda x_1, \lambda x_2, \dots, \lambda x_n) &= a_1 \lambda x_1 + \dots + a_n \lambda x_n \\ &= \lambda(a_1 x_1 + \dots + a_n x_n) \\ &= \lambda u(x_1, x_2, \dots, x_n) \end{aligned}$$

If $a_0 \neq 0$, u is not homogeneous, but it is still homothetic. To see this, notice that $h(x_1, \dots, x_n) = a_1 x_1 + \dots + a_n x_n$ is homogeneous as seen above, and define strictly increasing function $g(x) = a_0 + x$. Then $u = g \circ h$, implying that u is homothetic.

4. (15 points) Given a normed vector space $(V, +, \cdot, \|\cdot\|)$, show that for any $x \in V$, any open ball centered at x (with respect to the metric induced by the norm) is a convex set.

Answer: Take any $x \in V$. An open ball of radius r centered at x is defined as $B_r(x) := \{y \in X : d(x, y) = \|y - x\| < r\}$.

WTS: This set is convex, i.e.

$$\forall y_1, y_2 \in B_r(x), t \in [0, 1], ty_1 + (1-t)y_2 \in B_r(x)$$

or equivalently,

$$\forall t \in [0, 1], \|y_1 - x\| < r, \|y_2 - x\| < r \implies \|ty_1 + (1-t)y_2 - x\| < r$$

Observe that

$$\|ty_1 + (1-t)y_2 - x\| = \|t(y_1 - x) + (1-t)(y_2 - x)\| \leq t\|y_1 - x\| + (1-t)\|y_2 - x\| < tr + (1-t)r = r$$

5. (15 points) Let $f : X \rightarrow \mathbb{R}$ be a continuous real function on a metric space (X, d) . Let $Z(f) := \{p \in X : f(p) = 0\}$ be the set of all points in X at which $f(p) = 0$. Prove that $Z(f)$ is closed.

(Hint: You may want to use the sequential characterizations of closed sets and continuous functions as well as the fact that constant sequences converge to the constant itself.)

Answer: WTS: $Z(f)$ is closed. We use the sequential characterization of closedness, i.e. we want to show that for each sequence (p_n) in $Z(f)$ s.t. $p_n \rightarrow p \in X$, we have $p \in Z(f)$.

Notice first that by $p_n \in Z(f) \forall n$, we have $f(p_n) = 0 \forall n$. Given that $f(p_n) = 0 \forall n$, i.e. it is a constant sequence, we have $f(p_n) \rightarrow 0$. Since f is continuous, we have that $p_n \rightarrow p$ implies $f(p_n) \rightarrow f(p)$. Hence, by $f(p_n) \rightarrow 0$, $f(p_n) \rightarrow f(p)$, and the uniqueness of limits, we have $f(p) = 0$, which in turn implies $p \in Z(f)$. This completes the proof.

6. (10 points) Solve the integral

$$\int_2^3 x e^{x+1} dx.$$

Show your steps.

Answer: Notice first that this integral can be rewritten as

$$e \cdot \int_2^3 x e^x dx.$$

We then have

$$\begin{aligned} e \cdot \int_2^3 x e^x dx &= e \cdot \int_2^3 x d e^x \\ &= e \cdot \left([x e^x]_2^3 - \int_2^3 e^x dx \right) \\ &= e \cdot \left([x e^x]_2^3 - [e^x]_2^3 \right) \\ &= e \cdot (3e^3 - 2e^2 - (e^3 - e^2)) \\ &= e \cdot (2e^3 - e^2) \\ &= 2e^4 - e^3 \approx 89.11 \end{aligned}$$

where the first equality follows from the differentiation rule and the second equality follows from integration by parts.

7. (20 points) Consider the following constrained optimization problem:

$$\begin{aligned} \max_{(x,y) \in \mathbb{R}_{++}^2} \quad & \ln(x+1) + \ln(y+1) \\ \text{s.t.} \quad & x + 2y \leq c \\ & x + y \leq 2 \end{aligned}$$

where c is a positive constant.

(a) Write down the Lagrangian, and the necessary Kuhn-Tucker conditions for a point (x, y) to be a solution of the problem.

(b) Solve the problem for $c = \frac{5}{2}$.

(c) Let $V(c)$ denote the value function. Find the value of $V'(\frac{5}{2})$ using the Envelope Theorem.

Answer:

(a)

$$\mathcal{L}(x, y, \lambda_1, \lambda_2) = \ln(x+1) + \ln(y+1) + \lambda_1(c - x - 2y) + \lambda_2(2 - x - y)$$

FOC:

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{1}{x+1} - \lambda_1 - \lambda_2 = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{1}{y+1} - 2\lambda_1 - \lambda_2 = 0$$

CSC:

$$\lambda_1(c - x - 2y) = 0, \lambda_1 \geq 0, c - x - 2y \geq 0$$

$$\lambda_2(2 - x - y) = 0, \lambda_2 \geq 0, 2 - x - y \geq 0$$

(b) Let $c = \frac{5}{2}$. There are 4 cases:

Case 1: both slack

This contradicts the first FOC. Therefore no K-T solutions.

Case 2: only the first constraint binds

$\lambda_2 = 0$. Then $x + 2y = \frac{5}{2}$. Combining the first two FOCs we get $x + 1 = 2y + 2$. So $x = \frac{7}{4}, y = \frac{3}{8}$. But then $x + y = \frac{17}{8} > 2$, contradicting the second constraint. Therefore no K-T solutions.

Case 3: only the second constraint binds

$\lambda_1 = 0$. $x + y = 2$. Combining the first two FOCs, we get $x = y = 1$. But then $x + 2y = 3 > \frac{5}{2}$, violating the second constraint. Therefore no K-T solutions.

Case 4: both bind

$x + 2y = \frac{5}{2}$ and $x + y = 2$. Thus, $x = \frac{3}{2}, y = \frac{1}{2}$. Combining the first two FOCs we get $\lambda_1 = \frac{4}{15} \geq 0, \lambda_2 = \frac{2}{15} \geq 0$. Hence, $(x, y) = (\frac{3}{2}, \frac{1}{2})$ is the only solution candidate.

Since f is continuous and the constraint set is compact, there exists a solution. Therefore, $(x^*, y^*) = (\frac{3}{2}, \frac{1}{2})$ is the maximizer.

(c) By the Envelope Theorem, $V'(\frac{5}{2}) = \frac{d}{dc} \mathcal{L}(x^*, y^*, \lambda_1^*, \lambda_2^*, c) = \lambda_1^* = \frac{4}{15}$.

8. (15 points) Consider the correspondence $\Gamma : \mathbb{R} \rightrightarrows \mathbb{R}$

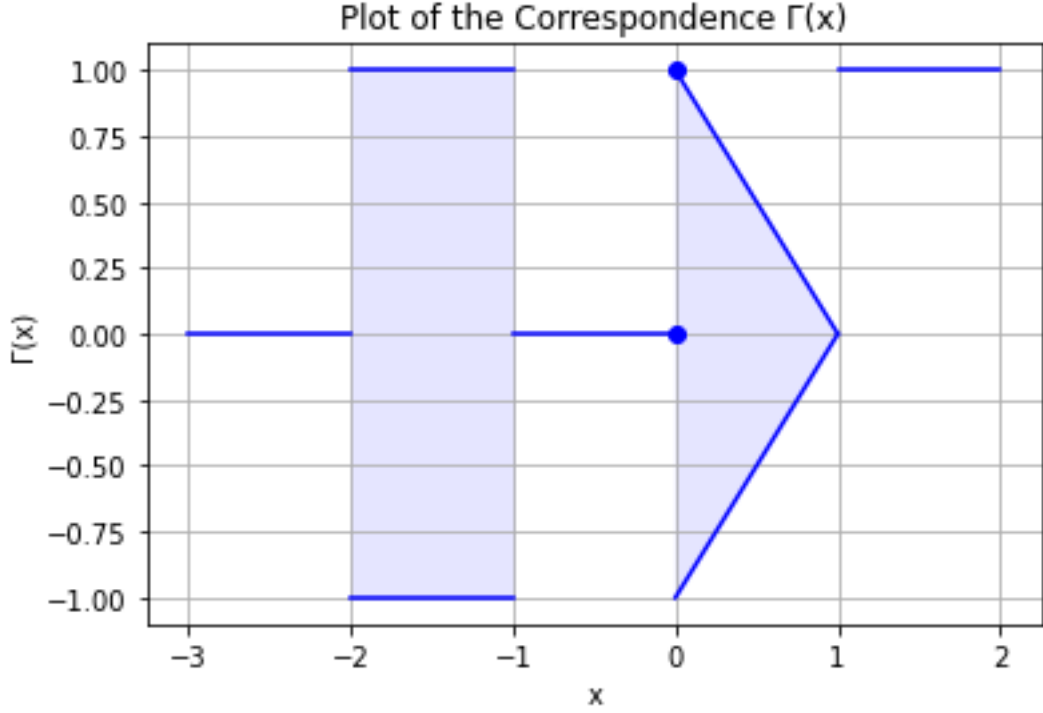
$$\Gamma(x) = \begin{cases} \{0\} & \text{if } x \in (-\infty, -2] \\ [-1, 1] & \text{if } x \in (-2, -1] \\ \{0\} & \text{if } x \in (-1, 0) \\ [0, 1] & \text{if } x = 0 \\ [-1 + x, 1 - x] & \text{if } x \in (0, 1) \\ \{1\} & \text{if } x \in [1, \infty) \end{cases}$$

(a) Plot the correspondence.

(b) Study the upper and lower hemicontinuity of Γ , i.e. state at which points the correspondence is upper hemicontinuous and at which points it is lower hemicontinuous. (No proof required)

Answer:

(a)



(b) $\Gamma(\cdot)$ is upper hemi-continuous on

$$\mathbb{R} \setminus \{-2, 0, 1\}$$

and lower hemi-continuous on

$$\mathbb{R} \setminus \{-1, 0, 1\}.$$

9. **Bonus Question** (10 extra points)

Prove the following statement:

If two finite subsets S and T of the usual inner product space $\mathbb{R}^n$ are strictly separated by a hyperplane (i.e. there exists some $p \in \mathbb{R}^n \setminus \{0\}$ and $c \in \mathbb{R}$ s.t. $S \subset \{x \in \mathbb{R}^n : p \cdot x < c\}$ and $T \subset \{x \in \mathbb{R}^n : p \cdot x > c\}$), then the convex hulls of S and T are disjoint.

Answer: We know that the convex hull of any finite set is equal to all convex combinations of its elements. Hence, for any $y \in \text{co}(S)$, we can find some $\{x_1, \dots, x_n\}$ in S and nonnegative $\{\lambda_1, \dots, \lambda_n\}$ s.t. $\sum_{i=1}^n \lambda_i = 1$ and $y = \sum_{i=1}^n \lambda_i x_i$. Given that $p \cdot x_i < c$ holds for each $i \in \{1, \dots, n\}$, we have $p \cdot y = p \cdot \sum_{i=1}^n \lambda_i x_i = \sum_{i=1}^n \lambda_i (p \cdot x_i) < \sum_{i=1}^n \lambda_i c = c \sum_{i=1}^n \lambda_i = c$. Likewise, for each $z \in \text{co}(T)$, we have $p \cdot z > c$. Summarizing, given $p \cdot y < c \quad \forall y \in \text{co}(S)$ and $p \cdot z > c \quad \forall z \in \text{co}(T)$, no element $w \in \mathbb{R}^n$ can be both in $\text{co}(S)$ and $\text{co}(T)$, meaning that the two convex hulls are indeed disjoint.